AM5920: Deep learning for Mechanics

Midsem

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AM25M009

Problem 1

$$f = \sin(100wx) + \sin(00wx) + \sin(1wx) + \text{gausian noise}$$
$$w = 2.5$$

Take a few number of points and fit a deep neural network with a relatively large number of neurons, by appropriate choice of hidden layers and neurons per layer. Investigate the issue of overfitting and how it can be handled with L1 and L2 regularizers on the neural network parameters. Between, L1 and L2 regularizers, which one is better?

model was chosen

- width = 10
- depth = 5
- Learning rate = 0.001

For this equation, model was trained for 50 epochs with

- No regularizer
- L1 regularizer
- L2 regularizer

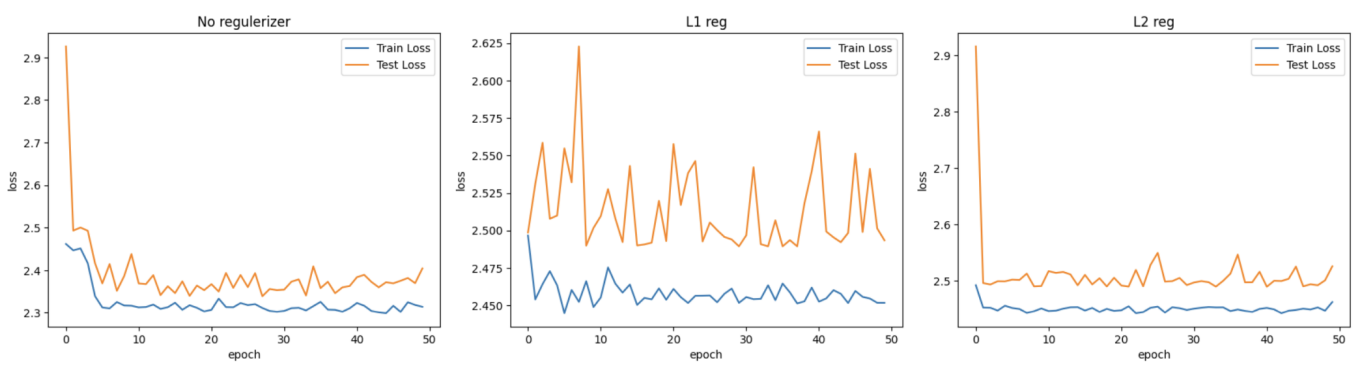

Figure 1: Different regularizer and effect in Loss

- **No Regularizer (left):** This is the clearest result. The training loss sits consistently below the test loss with a visible and stable gap, this indicates genuine overfitting. The model has effectively memorized the training points and fails to generalize to unseen data.
- **L1 Regularizer (middle):** The test loss is extremely noisy and unstable, spiking erratically throughout training. The regularization parameter λ is likely too aggressive, causing the model to struggle to find a stable solution. The train–test gap is actually larger here than without regularization, suggesting that L1 regularization is hurting performance rather than helping at this strength.

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- **L2 Regularizer (right):** This produces the smoothest result among the three. The training and test losses track each other closely with a minimal gap, indicating better generalization. Training also remains stable throughout.

  A small dataset was used to deliberately induce overfitting, visible in the no-regularizer case as a persistent gap between train and test loss. L2 regularization effectively reduced this gap by penalizing large weights smoothly, leading to a more generalizable model. L1 regularization, while theoretically capable of producing sparse solutions, introduced instability at the chosen lambda value. Overall, L2 is the better regularizer for deep neural networks, it provides smooth, stable penalization without the erratic gradient behavior that L1 can introduce near zero.

For the save of curiosity, here is a visualization of parameters being shrunk down

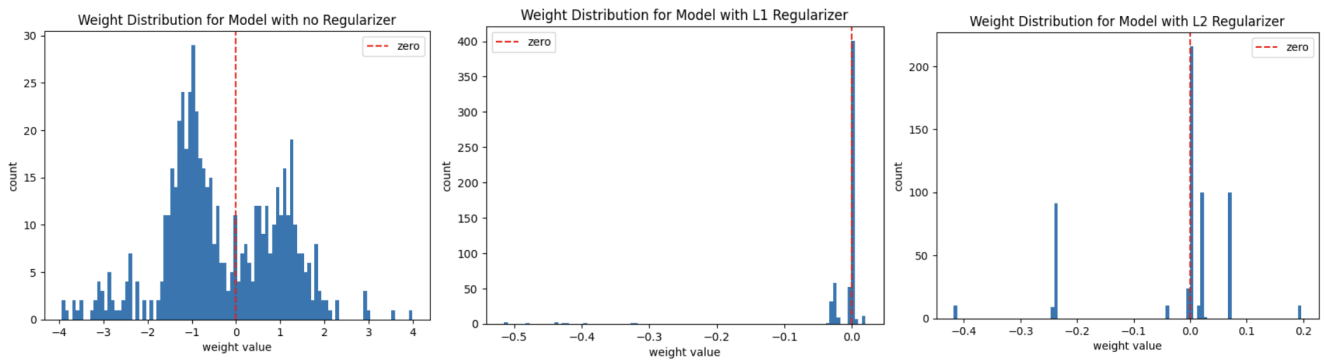

Figure 2: Visualization of parameters being ”reduced”

Problem 2

Comment on the generalization error for the above deep neural network. Does regularized loss function generalize better? Do we still need regularizer if we increase the number of data points that has been used to fit a deep neural network?

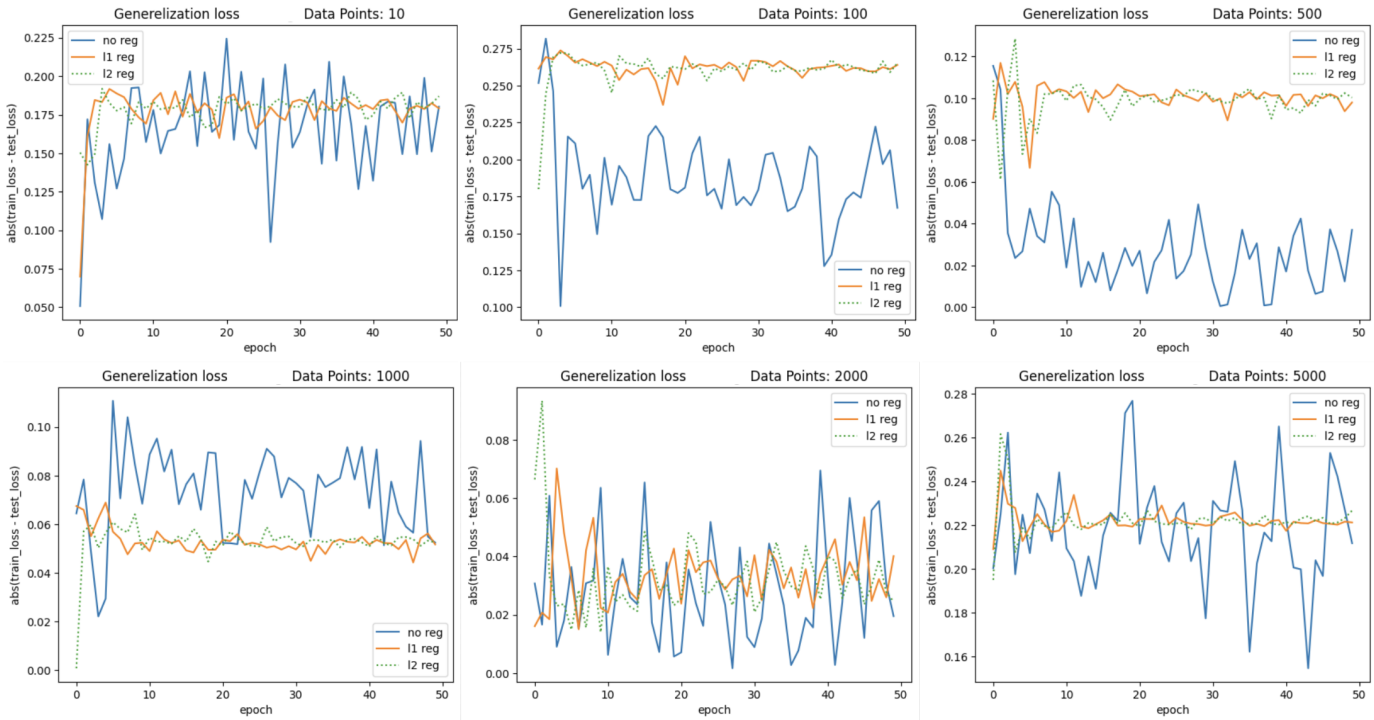

Figure 3: Dataset size vs regularizer

- $N = 10$ **(top left):** With only ten data points, all three curves look nearly identical and quite noisy. The dataset is simply too small for the model to learn any meaningful pattern. As a result, regularization has almost no visible effect, since there is very little structure in the data to constrain.

- $N = 100$ **(top middle):** This case produces one of the more interesting results. The model without regularization (blue) actually shows a smaller generalization gap compared to the L1 and L2 versions. Although this may seem counterintuitive, it is likely because the model has not fully overfit within the 50 training epochs. In this regime, the additional constraints imposed by regularization may slightly hinder learning. With more training epochs, the benefits of regularization would likely become more apparent.

- $N = 500$ **(top right):** A clearer trend begins to appear. The unregularized model (blue) exhibits a more variable generalization gap, while both L1 and L2 regularization settle toward a stable value around 0.1. Even if the gap is not always smaller, the regularized models behave more consistently during training.

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- $N = 1000$ **(bottom left):** This subplot shows the most obvious effect of regularization. The unregularized model fluctuates considerably, with the gap ranging roughly between 0.02 and 0.11. In contrast, both L1 and L2 regularization stabilize the gap near 0.05, indicating more reliable generalization.

- $N = 2000$ **and** $N = 5000$ **(bottom middle and right):** As the dataset size increases further, the differences between the three approaches begin to shrink. All curves gradually converge toward similar values. This suggests that with enough data, the model generalizes well even without explicit regularization, since the larger dataset itself provides a natural regularizing effect.

The generalization error, measured as the absolute difference between train and test loss, generally decreases as the number of training points increases, confirming that larger datasets naturally reduce overfitting. Regularization (L1 and L2) most visibly helps in the mid-range dataset sizes (N=500–1000), where it stabilizes the generalization gap and reduces variance. At very small N the dataset is too noisy for regularization to help, and at very large N the data itself acts as a regularizer, making explicit penalties less necessary. This suggests that regularization and dataset size are complementary, regularization matters most precisely when data is limited.

Problem 3

Investigate the spectral bias for the sigmoid activation function and how it gets mitigated for the ReLU activation function for the deep neural network.

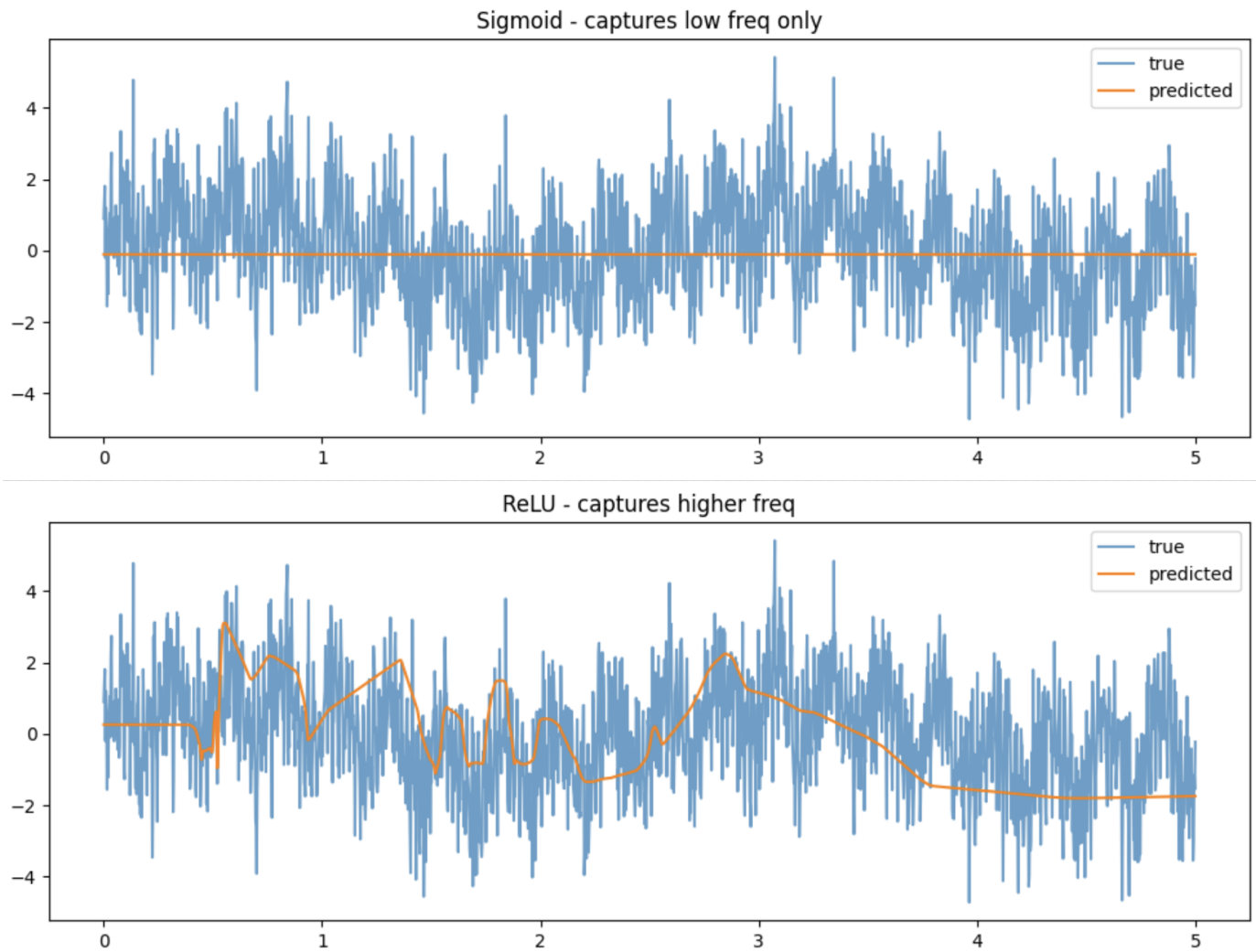

Figure 4: Spectral bias for ReLU vs Sigmoid

- **Sigmoid (top):** The predicted output is essentially a flat line at zero, indicating that the model has failed to learn any meaningful structure from the data. This appears to be an extreme case of spectral bias and vanishing gradients. As observed later in the layer-by-layer activation analysis, the sigmoid activations begin to saturate around layer 6, producing almost binary 0/1 outputs across the network. Once this saturation occurs, gradients become nearly zero, preventing the weights from updating. As a result, the model cannot learn any frequency component of the signal, whether low or high.

- **ReLU (bottom):** In contrast, the ReLU model shows clear learning behavior. The predicted

Problem 3 continued on next page. . .

curve follows the slow underlying wave of the function, tracking the broad envelope of the signal across the interval $[0, 5]$. It captures the low-frequency $\sin(wx)$ component reasonably well and even reflects some mid-frequency structure around $x \approx 0.5$ and $x \approx 3$. However, it completely misses the rapid oscillations of $\sin(100wx)$; the predicted curve remains smooth while the true signal fluctuates rapidly around it. This illustrates spectral bias in its classic form: neural networks tend to learn low-frequency patterns first and struggle to represent high-frequency components within a limited training budget.

Spectral bias refers to the tendency of neural networks to learn low frequency components of a target function first, with high frequencies learned slowly or not at all. For the sigmoid activation function, this bias is terrible, zeroing of neurons in deep layers causes vanishing gradients, completely preventing learning. ReLU takes care of this by allowing gradients to flow freely through positive activations, enabling the network to learn the low frequency envelope of the target function. However, the highest frequency component sin(100wx) remains largely uncaptured even with ReLU, demonstrating that spectral bias is reduced but not eliminated.

Problem 4

Investigate the spectral bias for the sigmoid activation function and how it gets mitigated for the ReLU activation function for the deep neural network.

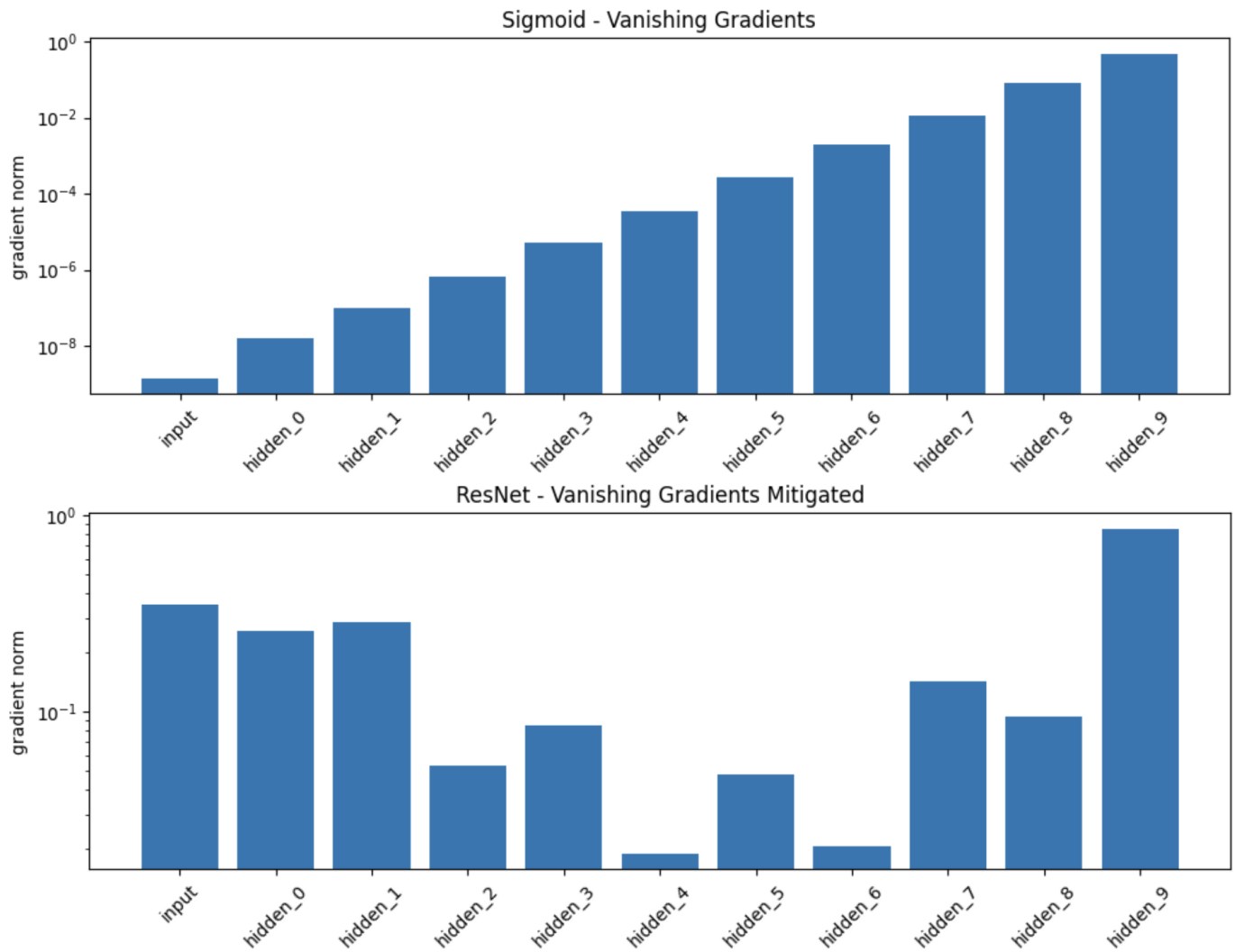

Figure 5: With and wothout resnet comparison

- **Sigmoid - Vanishing Gradients (top):** The pattern is striking and almost perfectly predictable. Starting from the input layer at approximately 10^{-9}, the gradient norm increases by roughly one order of magnitude at each layer, eventually reaching about 10^0 at `hidden_9`. This corresponds to a span of nearly nine orders of magnitude across only ten layers. In practical terms, the earliest layers receive almost no gradient signal, meaning their weights barely update during training. As a result, only the final few layers actually learn, and the network behaves much like a shallow model despite having ten layers.

  This exponential decay occurs because backpropagation repeatedly multiplies gradients by

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the derivative of the sigmoid function.

- **ResNet - Vanishing Gradients Mitigated (bottom):** The behavior is dramatically different in the residual network. Here, the gradient norms across all layers remain within the range of roughly 10^{-2} to 10^{0}, spanning only about two orders of magnitude. The input layer receives a gradient of around 0.3, which is comparable to the gradient at the output layer. Instead of decaying systematically with depth, the gradients fluctuate slightly but remain consistently strong throughout the network. This demonstrates how residual connections help preserve gradient flow, allowing even very deep networks to train effectively.

Problem 5

Investigate whether the spectral bias is less severe for the residual network than the deep neural network.

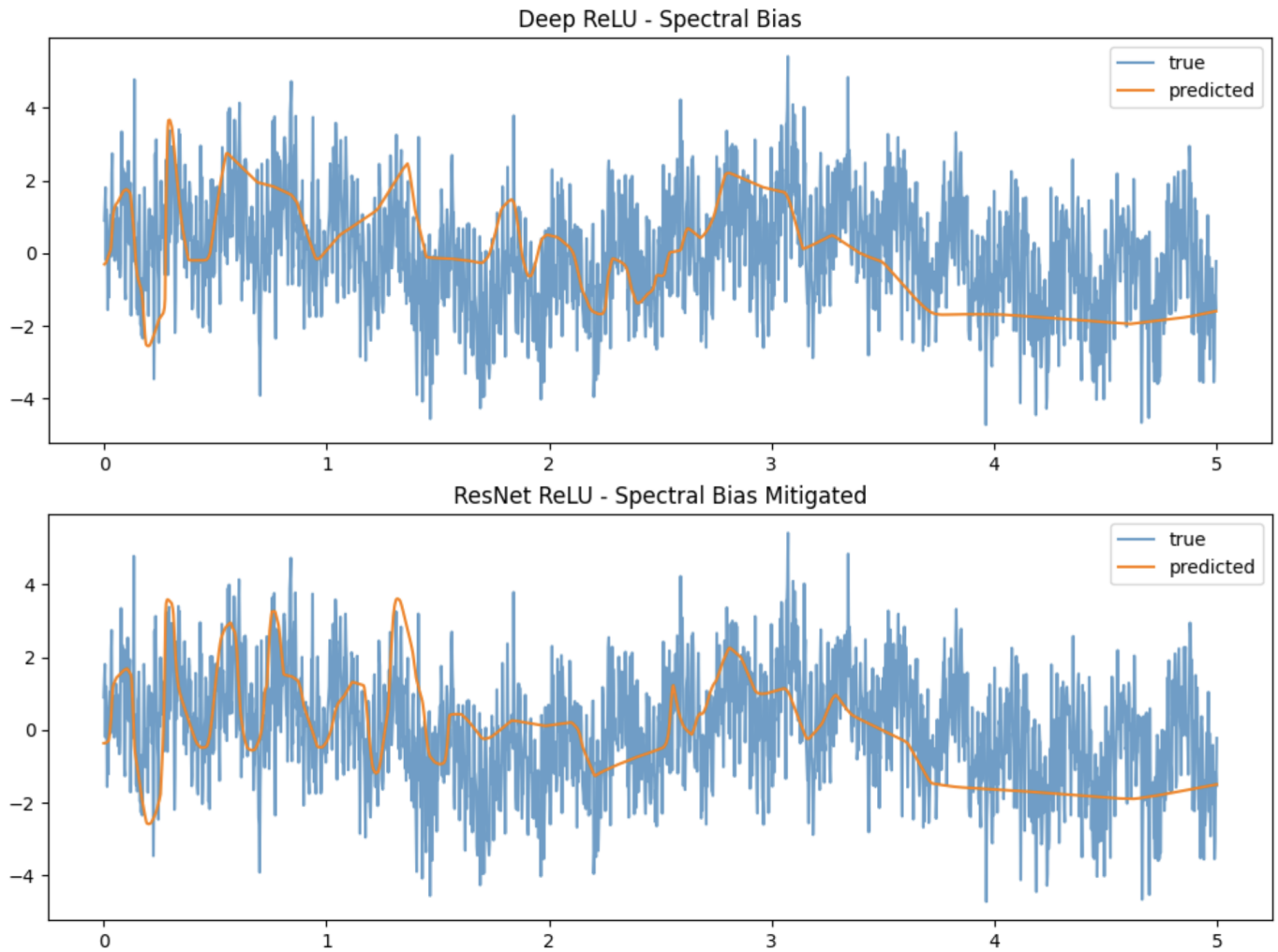

Figure 6: Spectral Bias with and without Resnet

- **Deep ReLU (top):** The model captures the broad, slow wave across the entire domain, but the predicted curve appears noticeably smooth. It follows the overall envelope of the signal but fails to reproduce the faster oscillations present in the true function. Beyond $x \approx 3$, the prediction becomes increasingly inaccurate and begins to lose even the mid-frequency structure. In effect, the model has mainly learned the low-frequency component $\sin(wx)$ and only partially captures $\sin(10wx)$ in the earlier part of the domain, with performance degrading toward the right side of the plot.

- **ResNet ReLU (bottom):** The ResNet produces a visibly richer prediction. The curve follows the true signal more closely across the entire domain, with more peaks and troughs that

Problem 5 continued on next page. . .

correspond to the actual oscillations of the function. In particular, the region between $x = 0$ and $x = 3$ shows clearer evidence that some faster oscillations are being captured compared to the standard deep ReLU model. Importantly, the ResNet also maintains better fidelity further along the domain, where the deep ReLU network had already started to break down.

Despite this improvement, both models still fail to capture the highest-frequency component $\sin(100wx)$. The true signal continues to oscillate much faster than either model can represent, leaving this component effectively invisible in the predictions.

Source Code

The complete implementation for this project is available at:

https://github.com/Mafaz03/Deep-Learning-in-Mechanics/blob/main/midsem.ipynb